CSCI 335 Software Design and Analysis III

Sorting I

Insertion sort, MergeSort, Heapsort, Quicksort, Quickselect, Introsort

Worst and average-case analysis

Visualizations

https://www.cs.usfca.edu/~galles/visualization/ComparisonSort.html

Sorting algorithms

- Comparison-based sorting
- STL versions:

```
sort(v.begin(), v.end() );
sort(v.begin(), v.begin() + (v.end()-v.begin())/2);
sort(v.begin(),v.end(),greater<int>{ });
```

Also: stable_sort()

- Does not swap the positions of two elements with the same comparison key.

Insertion Sort

- Pass p=1 through N-1:

  items in positions 0 through p are in sorted order.

Original	34	8	64	51	32	21	Positions Moved
After $p = 1$	8	34	64	51	32	21	1
After $p = 2$	8	34	64	51	32	21	0
After $p = 3$	8	34	51	64	32	21	1
After $p = 4$	8	32	34	51	64	21	3
After $p = 5$	8	21	32	34	51	64	4

Insertion Sort

```
template <typename Comparable>
void InsertionSort(vector<Comparable> & a) {
    for (int p = 1; p < a.size(); ++p) {
        Comparable tmp = std::move(a[p]);
        int j;
        for (j = p; j > 0 && tmp < a[j - 1]; --j)
            a[j] = std::move(a[j - 1]);
        a[j] = std::move(tmp);
    }
}
```

Stl implementation

```
// Two-parameter version calls three-parameter version.
// Uses C++11 decltype.
template <typename Iterator>
void InsertionSort(const Iterator &begin, const Iterator &end) {
  InsertionSort(begin, end, less<decltype(*begin)>{});
}
```

Stl implementation

```
// Three-parameter version.
template <typename Iterator, typename Comparator>
void InsertionSort(const Iterator &begin, const Iterator &end, Comparator
                                                                less_than) {
  if (begin == end) return;
  for (Iterator p = begin + 1; p != end; ++p) {
    auto tmp = std::move(*p);
    Iterator j;
    for (j = p; j != begin && less_than(tmp, *(j - 1)); --j)
        *j = std::move(*(j - 1));
    *j = std::move(tmp);
}
---
You can call for instance:
InsertionSort(vec.begin(), vec.end()); // vec is of type vector<string>
```

Insertion Sort Analysis

- Worst case: $\Theta(N^2)$
- Best case (presorted input): $O(N)$
  - One comparison per item

Lower bound for simple sorting algorithms

Inversion in an array **a** is an ordered pair (i, j), 0 <= i, j <**a**.size():

i < j and **a**[i] > **a**[j]

Example array: 34, 8, 64, 51, 32, 21

9 inversions: (34,8), (34,32), (34,21),
(64,51), (64,32), (64,21),
(51,32),(51,21),
(32,21)

These are exactly the swaps performed by insertion sort!

=> Swapping two adjacent items that are out of place reduces one inversion

=> Sorted array needs no inversions

Lower bound for simple sorting algorithms

- Average number of inversions in permutation of N integers (assume no duplicates) provides

Average running time for insertion sort and for **all algorithms that are based on swapping adjacent elements !**

Lower bound for simple sorting algorithms

- **Theorem:** Average number of inversions in an array of n distinct elements is n(n-1)/4
- **Proof:**

  Consider list L of n integers and its reverse Lr.

  Total number of inversions for both lists is ?

  Average number of inversions is ?
- **Therefore:** Any algorithm that sorts by exchanging adjacent elements is **$\Omega(n^2)$ on average**.

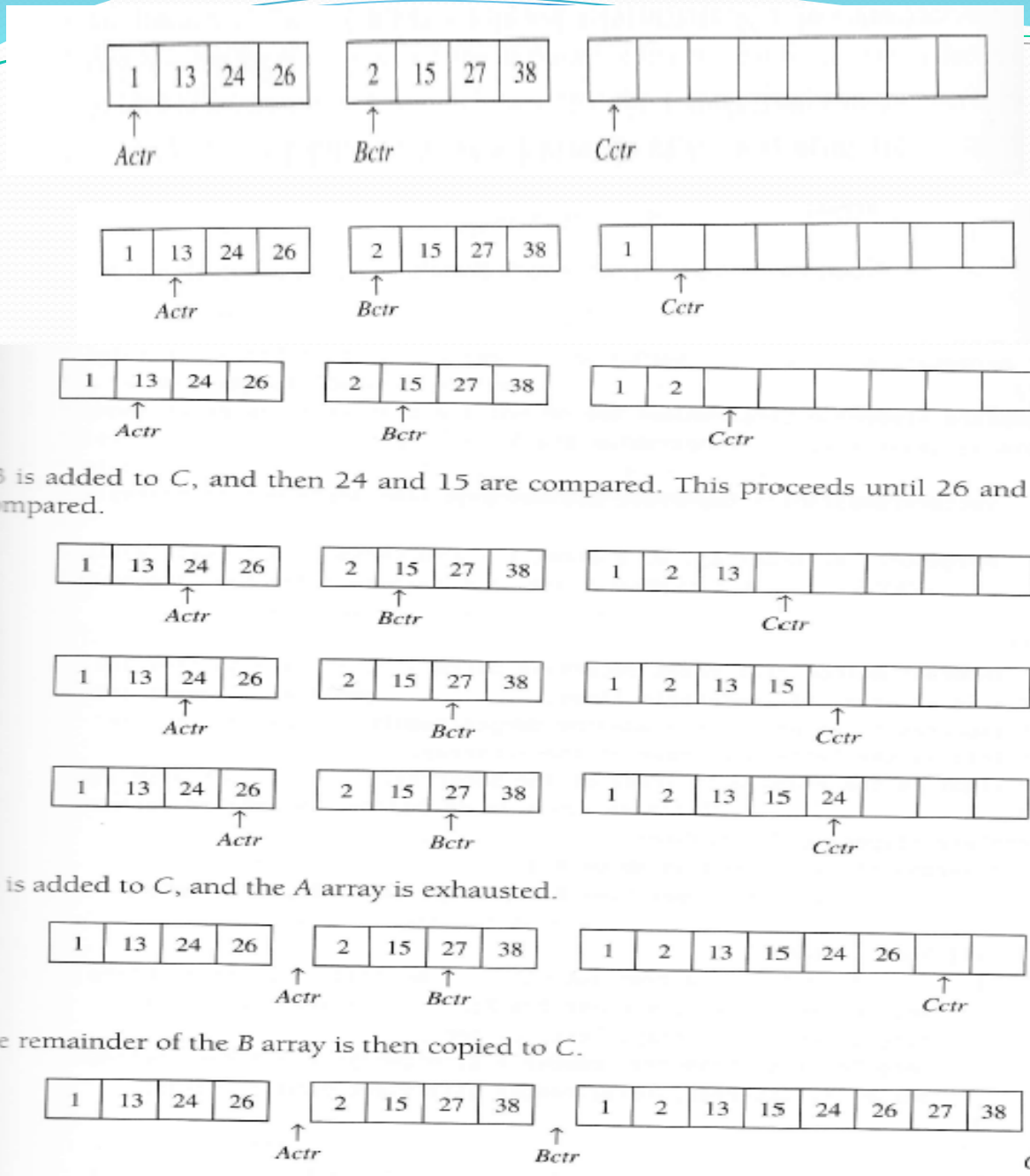
Lower Bound for Simple Sorting

- *Theorem*. Average number of inversions in an array of n distinct elements is $\frac{n(n-1)}{4}$.
- *Proof*. Consider list L of n integers and its reverse L_r. For every pair of indicies (i, j), either (i, j) is an inversion in L or in L_r. Therefore the total number of inversions in both lists is $\binom{n}{2}$. The average number of insertions across all lists is $\frac{n!}{2n!}\binom{n}{2} = \frac{n(n-1)}{4}$.
- Thus, any algorithm that sorts by exchanging adjacent elements is $\Omega(n^2)$ on average.

Mergesort

- O(N logN) worst-case running time
- Near optimal comparisons
- Recursive
- Key: O(N) for merging two sorted lists/arrays
- Need for extra O(N) space to merge arrays
- Good for linked lists, no need for random access
  - No need for extra space to merge sublists
- Stable sort
- May be better in C++11 with move semantics

Merge two sorted arrays (or lists)



3 is added to C, and then 24 and 15 are compared. This proceeds until 26 and 27 ar
mpared.

is added to C, and the A array is exhausted.

e remainder of the B array is then copied to C.

Figure 8.2
Top-down mergesort example

Each line shows the result of a call on `merge` *during top-down mergesort. First, we merge A and S to get A S; then, we merge O and R to get O R; then, we merge O R with A S to get A O R S. Later, we merge I T with G N to get G I N T, then merge this result with A O R S to get A G I N O R S T, and so on. The method recursively builds up small sorted files into larger ones.*

From, Algorithms in C++,
Robert Sedgewick

Mergesort Analysis

- $T(0) = T(1) = 1$
- $T(n) = 2T\left(\frac{n}{2}\right) + n$

We assume that n is a power of 2, i.e. n = 2^k for some positive k

- We will solve this recurrence relation by creating a telescoping sum. (The other way we've been doing it is also in the book)
- $\frac{T(n)}{n} = \frac{2T\left(\frac{n}{2}\right)}{n} + \frac{n}{n} = \frac{T(n/2)}{n/2} + 1$

Mergesort Analysis

Sort array of size n

$$\frac{T(n)}{n} = \frac{T(n/2)}{n/2} + 1$$

Sort array of size n/2

$$\frac{T(n/2)}{n/2} = \frac{T(n/4)}{n/4} + 1$$

Sort array of size n/4

$$\frac{T(n/4)}{n/4} = \frac{T(n/8)}{n/8} + 1$$

.

.

.

Sort array of size 2

$$\frac{T(2)}{2} = \frac{T(1)}{1} + 1$$

Mergesort Analysis

Sum of left sides:

$T(n)/n + T(n/2)/(n/2) + T(n/4)/(n/4) + \ldots + T(4)/4 + T(2)/2$

Equals

Sum of right sides:

$T(n/2)/(n/2) + 1 + T(n/4)/(n/4) + 1 + T(n/8)/(n/8) + 1 + \ldots + T(2)/2 + 1 + T(1)/1 + 1$

Mergesort Analysis

Sum of left sides:

$T(n)/n + T(n/2)/(n/2) + T(n/4)/(n/4) + \ldots + T(4)/4 + T(2)/2$

Equals

Sum of right sides:

$T(n/2)/(n/2) + 1 + T(n/4)/(n/4) + 1 + T(n/8)/(n/8) + 1 + \ldots + T(2)/2 + 1 + T(1)/1 + 1$

Equal parts cancel out

Mergesort Analysis

Sum of left sides:

T(n)/n

Equals

Sum of right sides:

1 + 1 + 1 + … + 1 + T(1)/1 + 1

How many 1's ?

Sum of left sides:

$T(n)/n$

Equals

Sum of right sides:

$1 + 1 + 1 + \ldots + 1 + T(1)/1 + 1$

How many 1's ? Answer: log(n)

[Hint: If n is a power of 2 how many times can we divide n until we reach 1?]

Mergesort Analysis

$T(n)/n = \log(n) + T(1)/1 \Rightarrow$

$T(n) = n \log(n) + n\,T(1) \Rightarrow$

$T(n) = n \log(n) + n \text{ [since } T(1) = 1] \Rightarrow$

$$\boldsymbol{T(n) = O(n \log(n))}$$

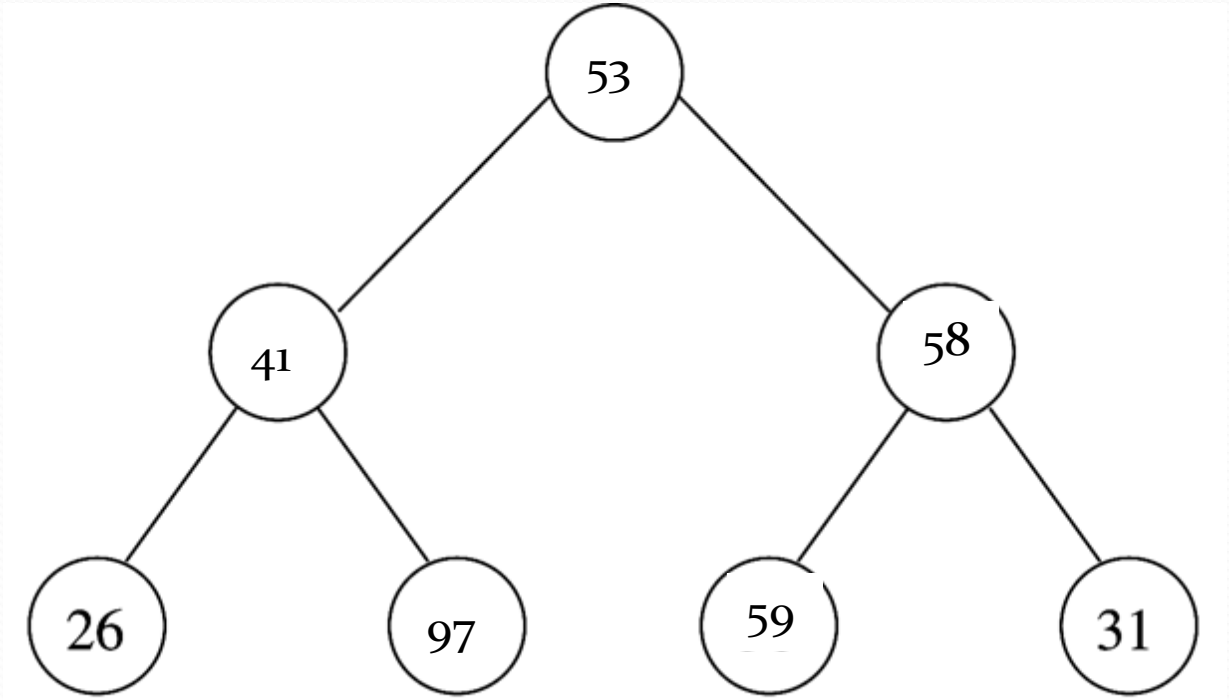
Heapsort

- Uses binary heap implemented as array
- O(N logN) time bound [worst and average]
- In-place algorithm, uses O(1) extra memory.
- Simple: Start with a given array:

    A) Build binary heap from N element: O(N)

    B) Perform deleteMax() N times => O(NlogN)

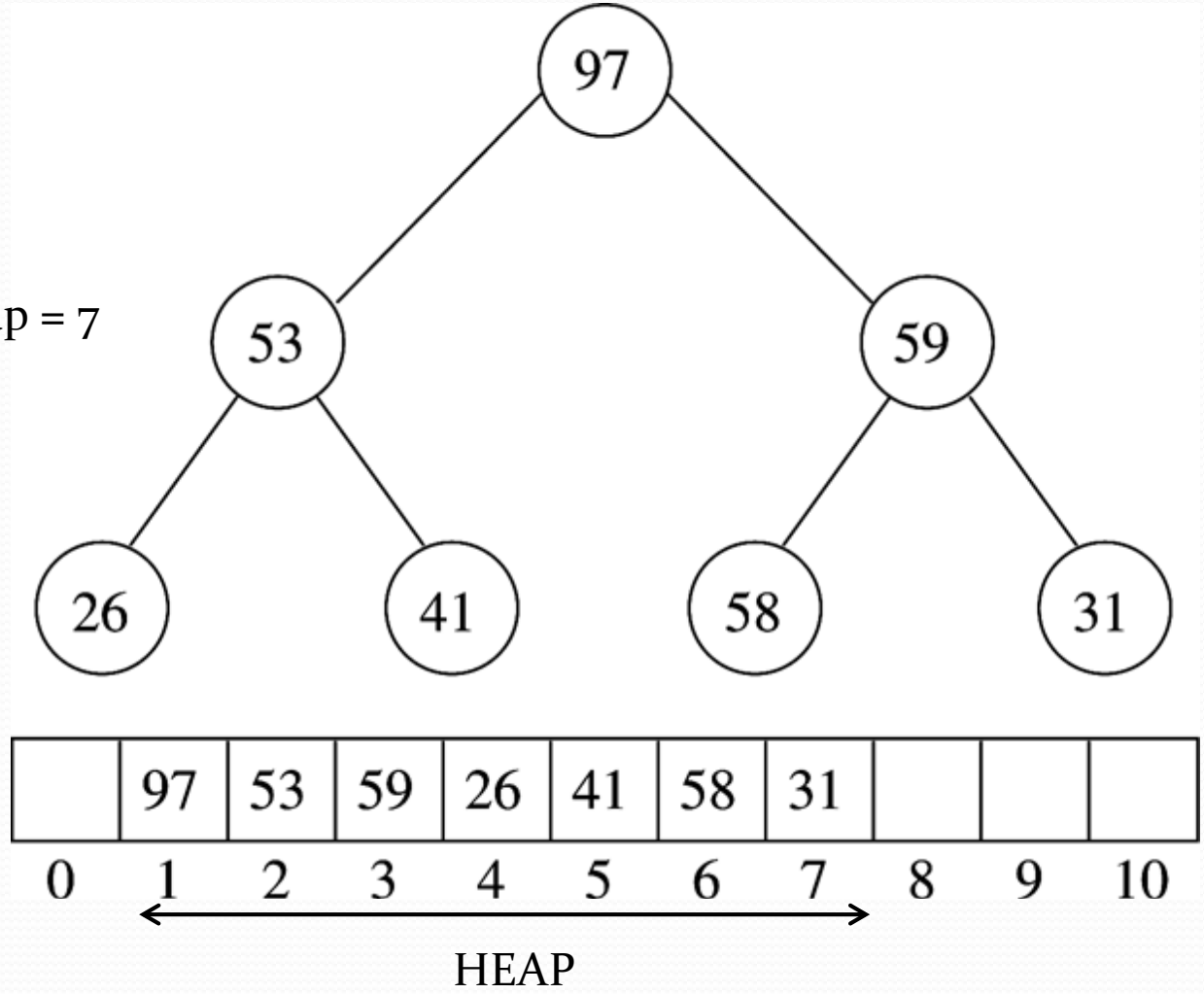
Original array



	53	41	58	26	97	59	31			
0	1	2	3	4	5	6	7	8	9	10

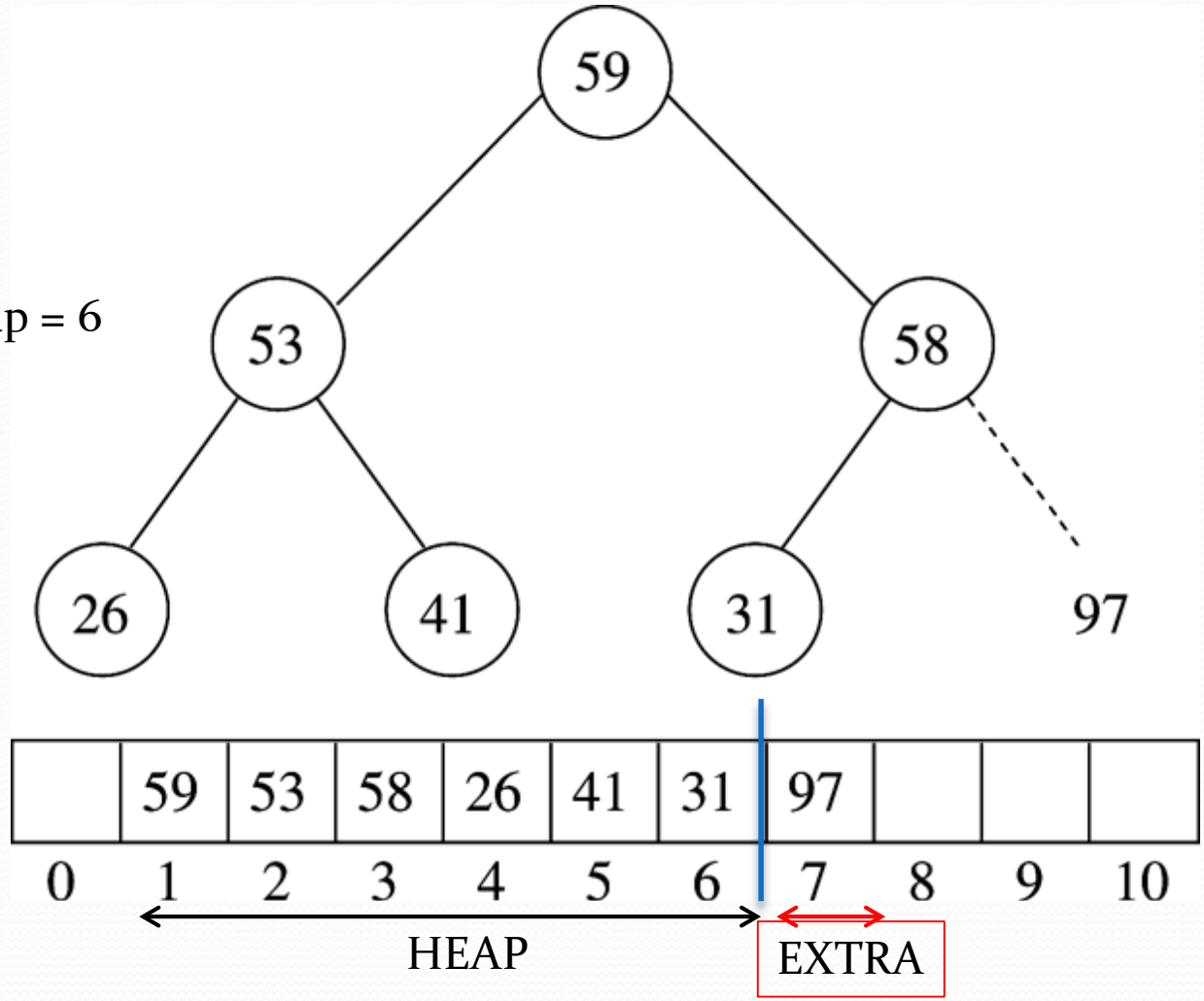
After build heap

size_of_heap = 7



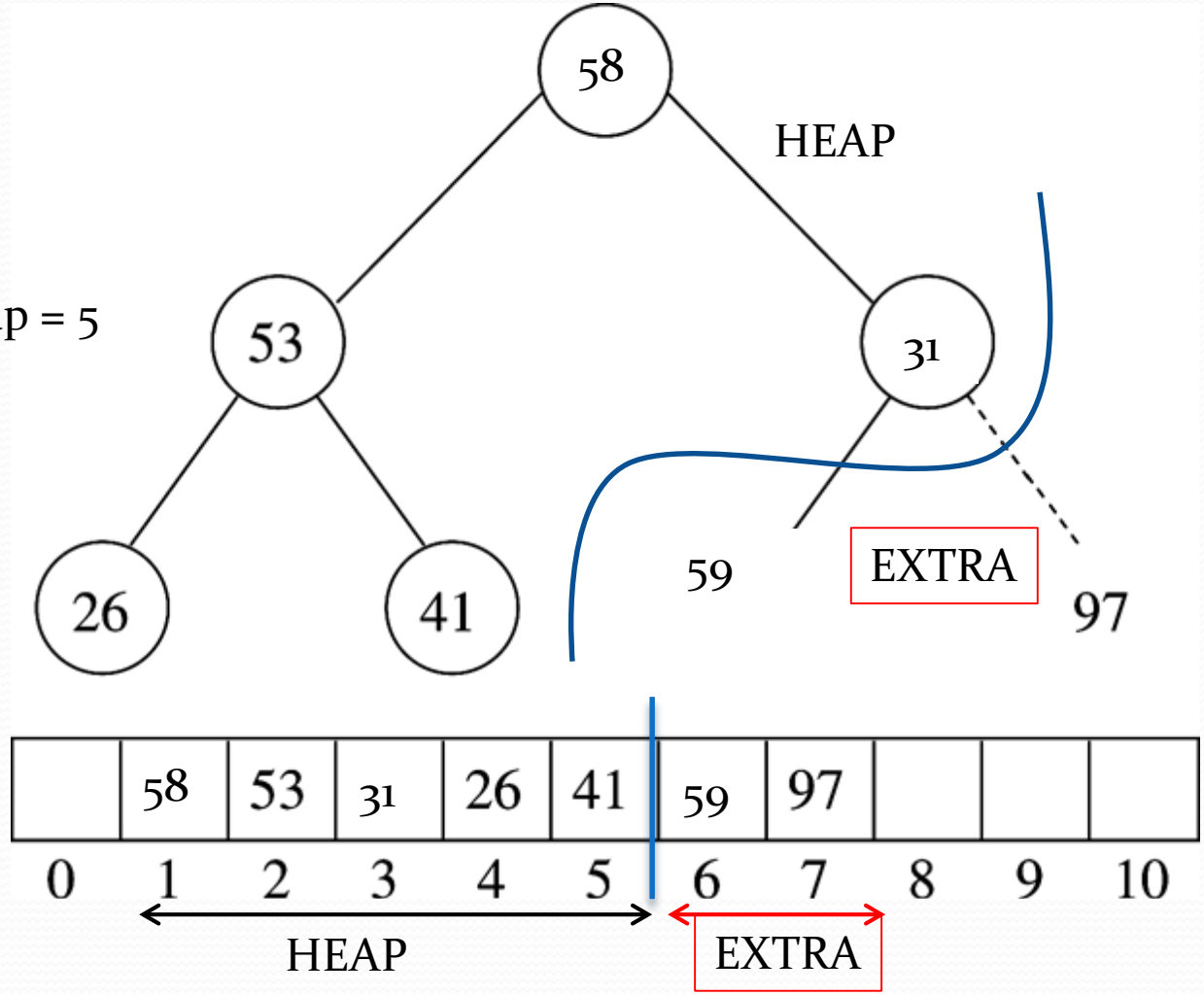
After first deleteMax()

size_of_heap = 6



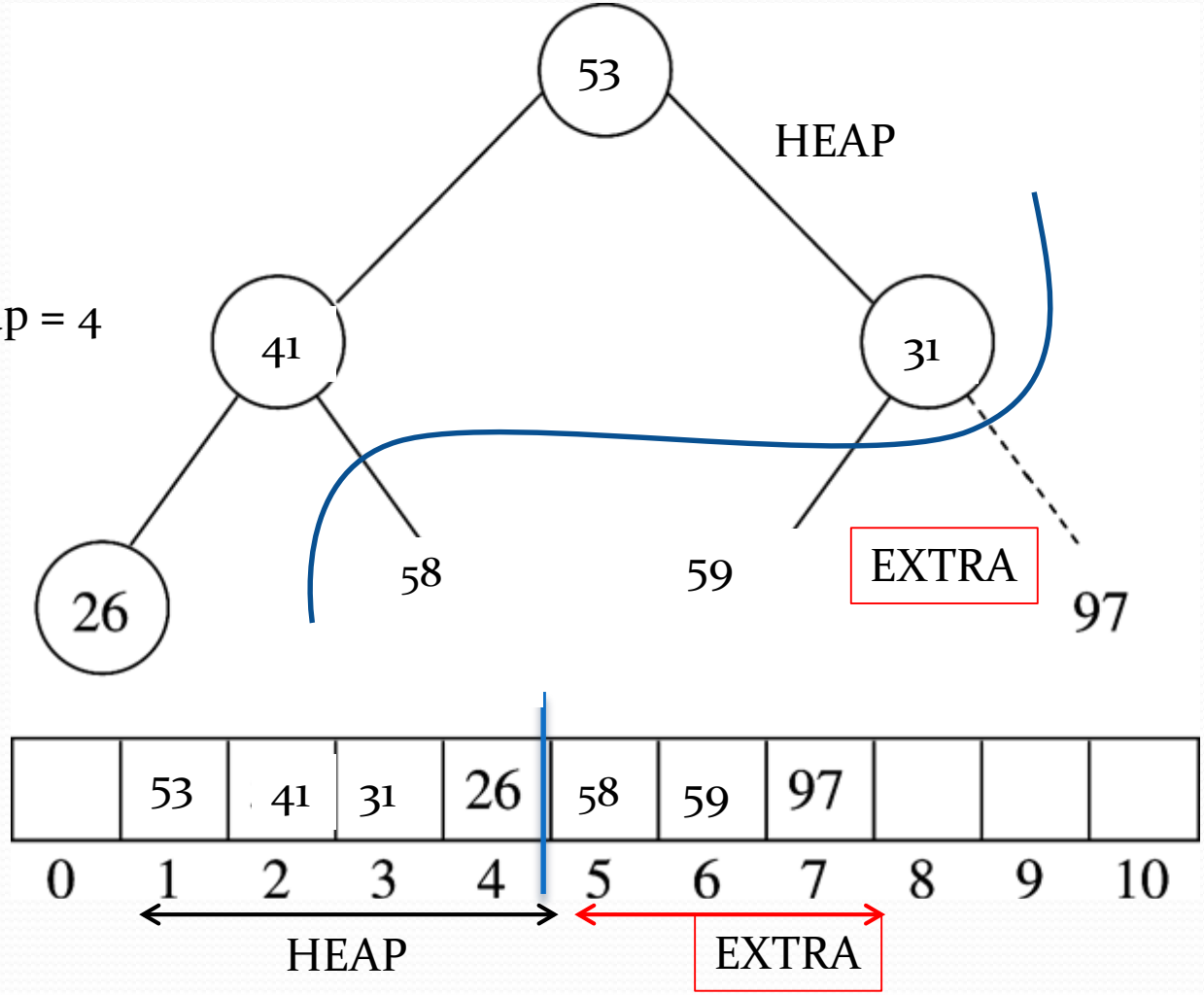
After second deleteMax()

size_of_heap = 5

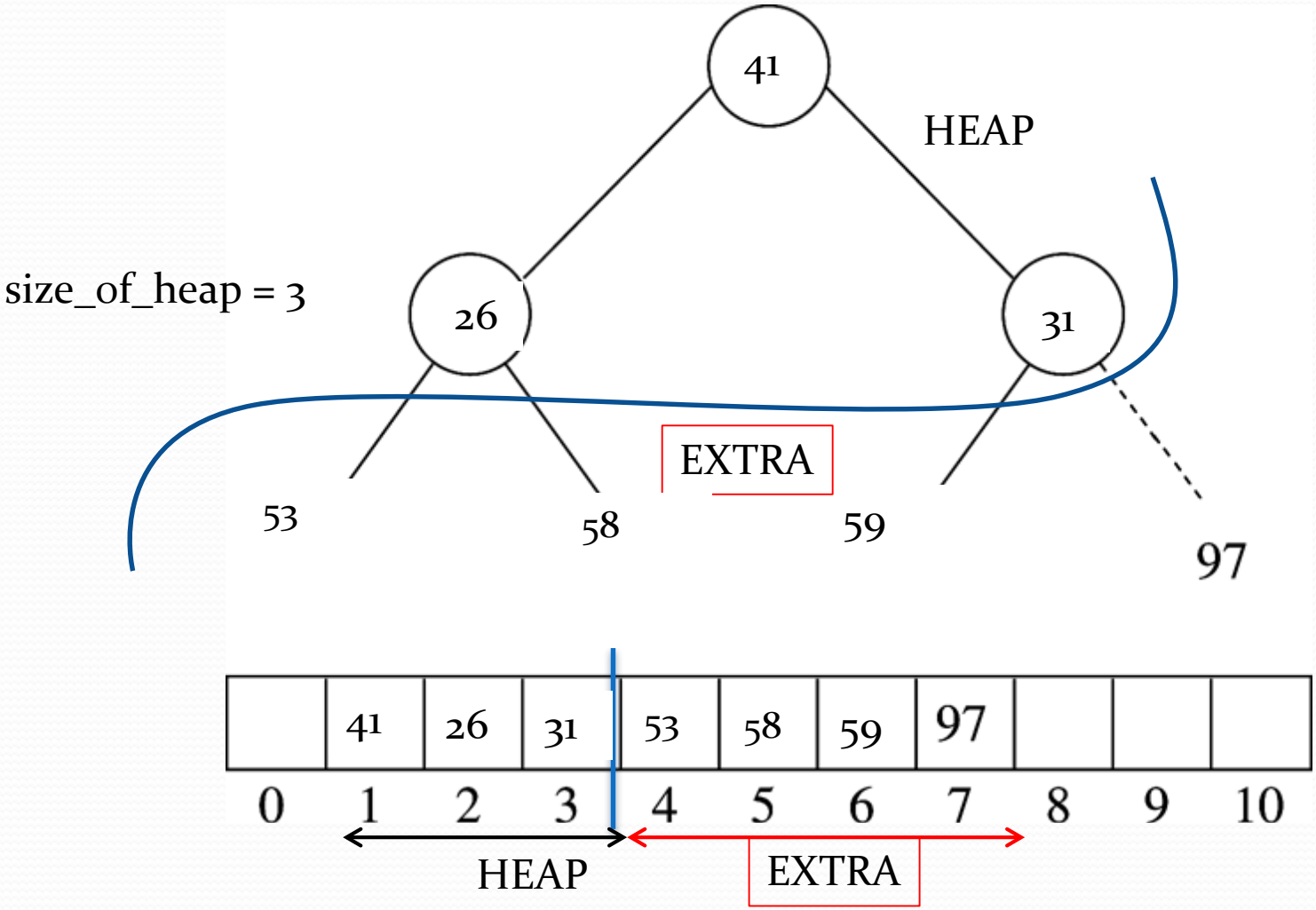


After third deleteMax()

size_of_heap = 4

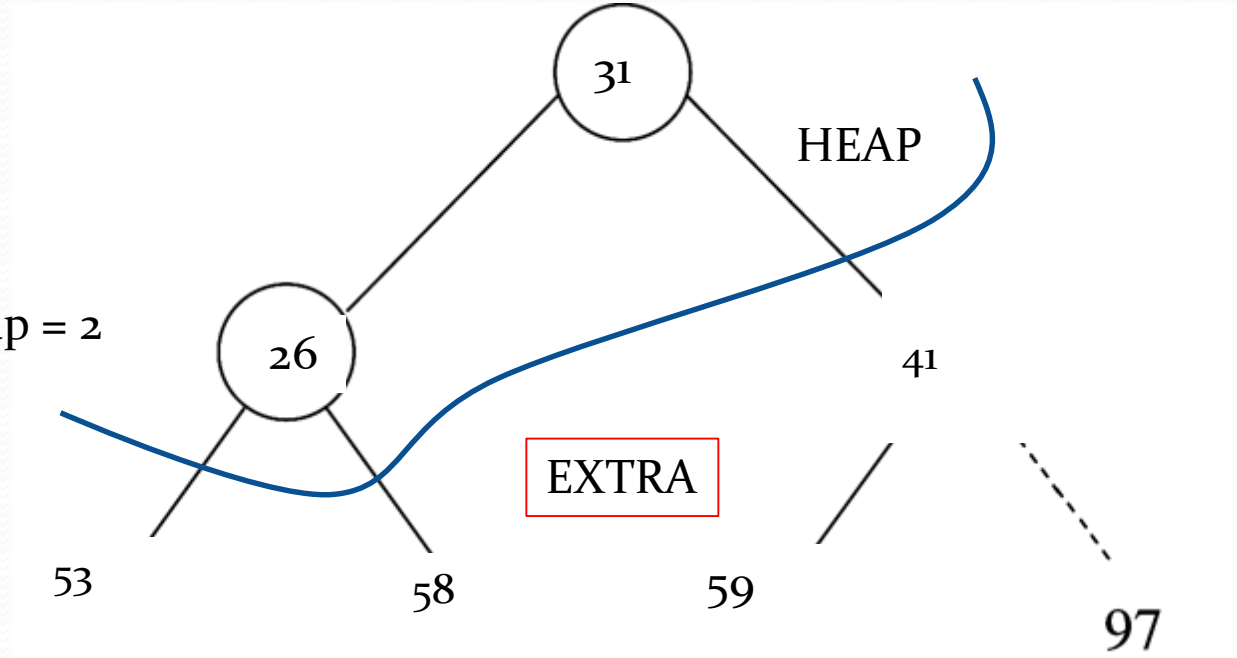


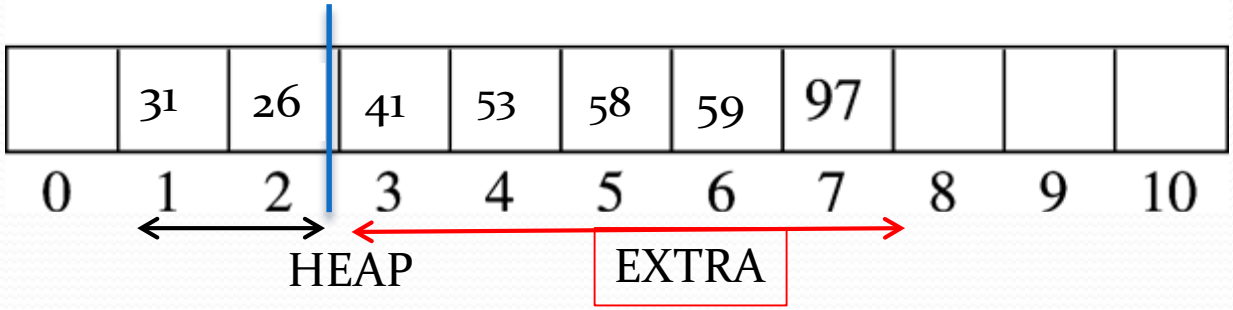
After next deleteMax()



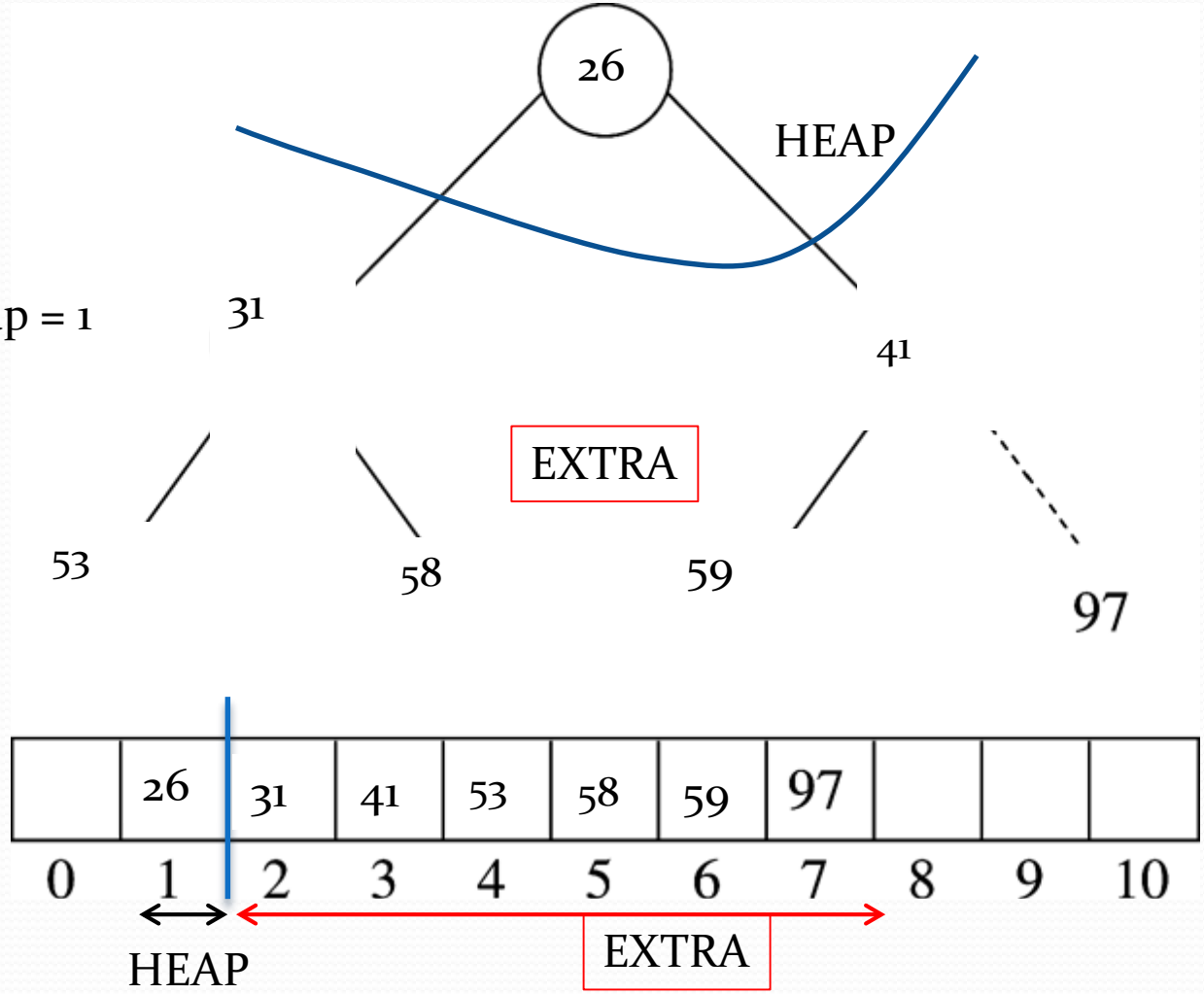
After next deleteMax()

size_of_heap = 2



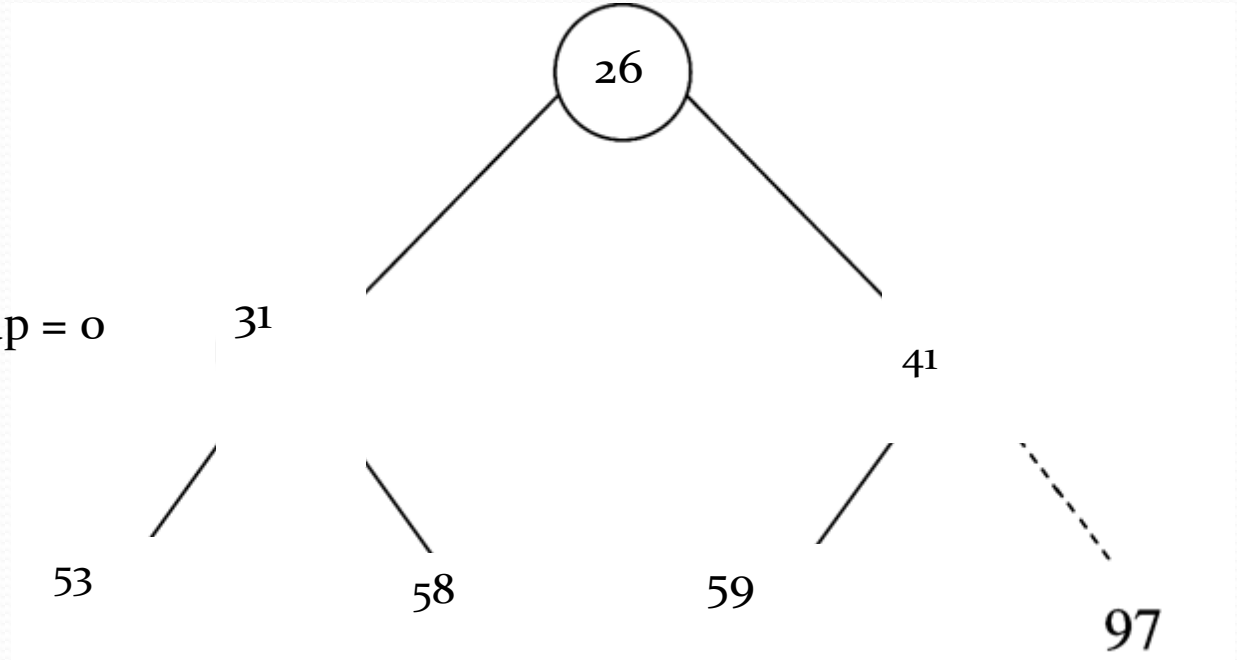


After next deleteMax()



Done!

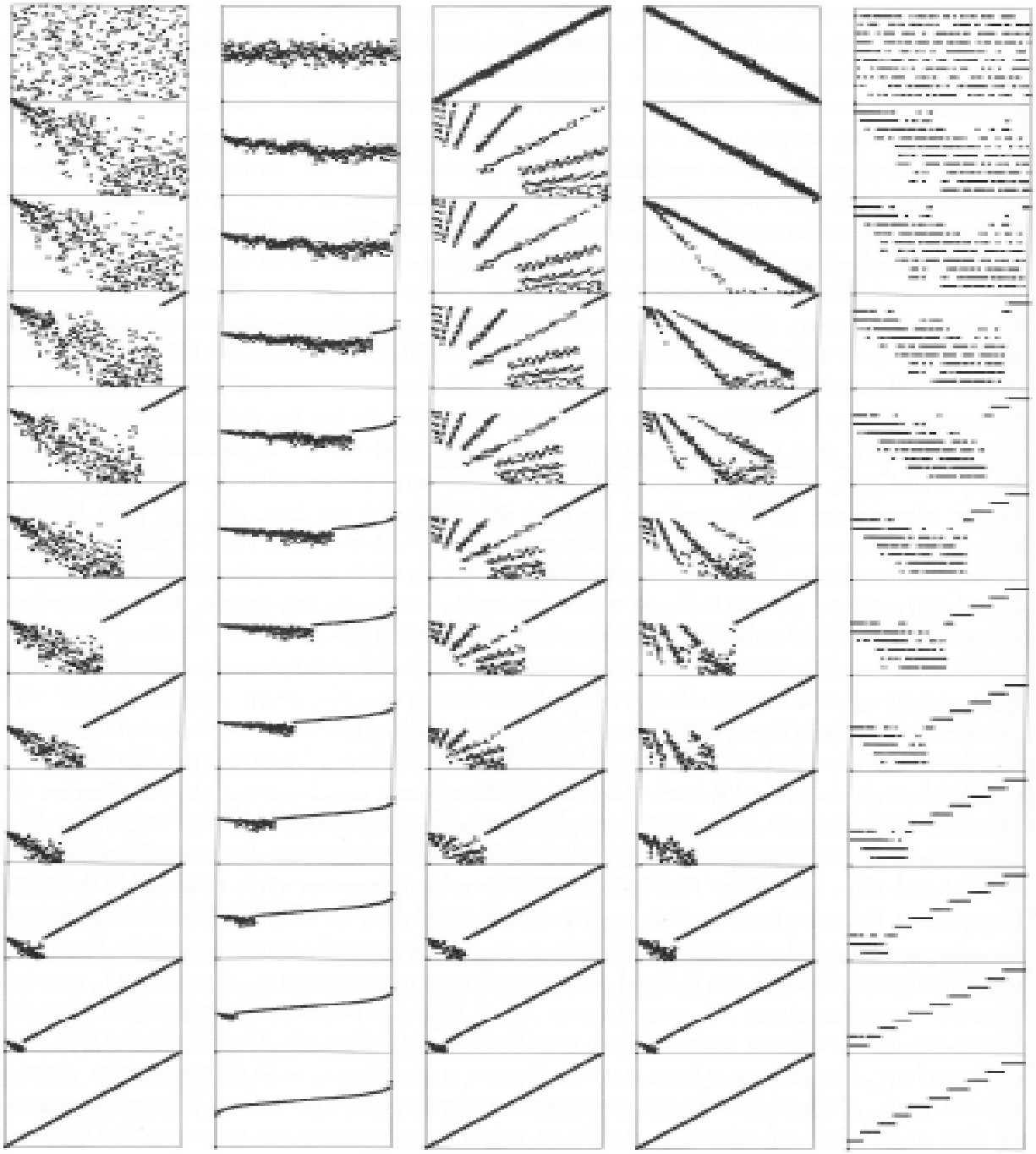
size_of_heap = 0



	26	31	41	53	58	59	97			
0	1	2	3	4	5	6	7	8	9	10

Figure 9.12
Dynamic characteristics of heapsort on various types of files

The running time for heapsort is not particularly sensitive to the input. No matter what the input values are, the largest element is always found in less than lg N steps. These diagrams show files that are random, Gaussian, nearly ordered, nearly reverse-ordered, and randomly ordered with 10 distinct key values (at the top, left to right). The second diagrams from the top show the heap constructed by the bottom-up algorithm, and the remaining diagrams show the sortdown process for each file. The heaps somewhat mirror the initial file at the beginning, but all become more like the heaps for a random file as the process continues.



From, Algorithms in C++, Robert Sedgewick

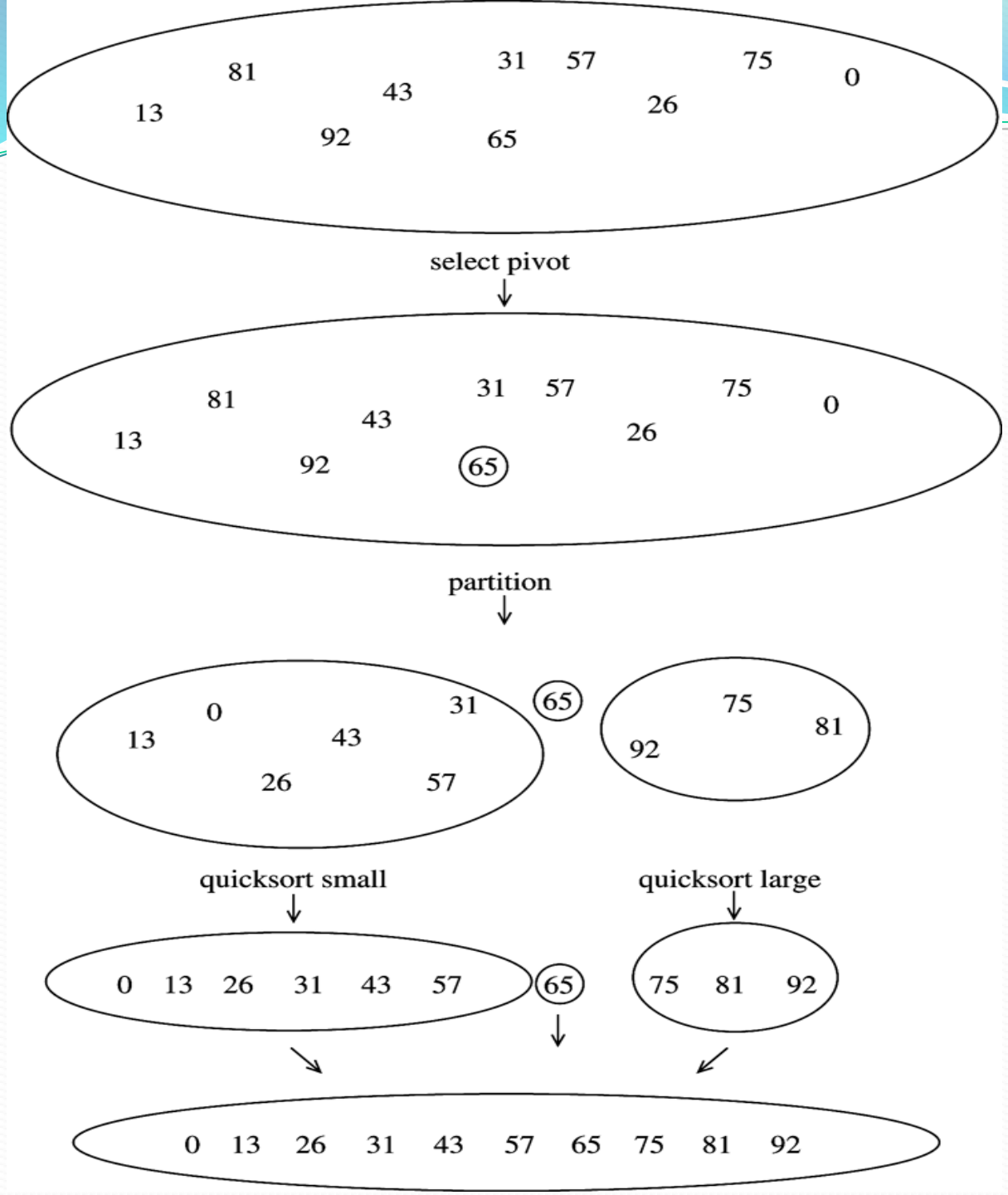
Quicksort

- Fastest for C++, more comparisons but fewer swaps
- O(NlogN) average, $O(N^2)$ worst
- To sort array S:

  quicksort(S)
  - If |S|=0, or 1 then return
  - **Pick** any element v in S. This is the **pivot**.
  - Partition S - {v} into two disjoint groups:

    $S_1=\{x \text{ in } S - \{v\} \mid x \leq v\}$ and $S_2=\{x \text{ in } S - \{v\} \mid x \geq v\}$
  - **Return** quicksort(S_1), v , quicksort(S_2)

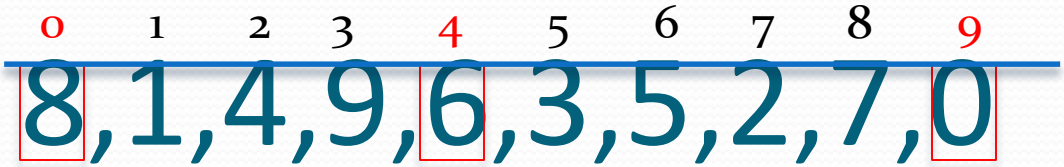
  Visualization:
  https://opendsa-server.cs.vt.edu/embed/quicksortAV


81 31 57 75 0
43
13 26
92 65
select pivot
81 31 57 75 0
43
13 26
92 65
partition
0 31 65 75
13 43 81
26 57 92
quicksort small
quicksort large
0 13 26 31 43 57 65 75 81 92
0 13 26 31 43 57 65 75 81 92

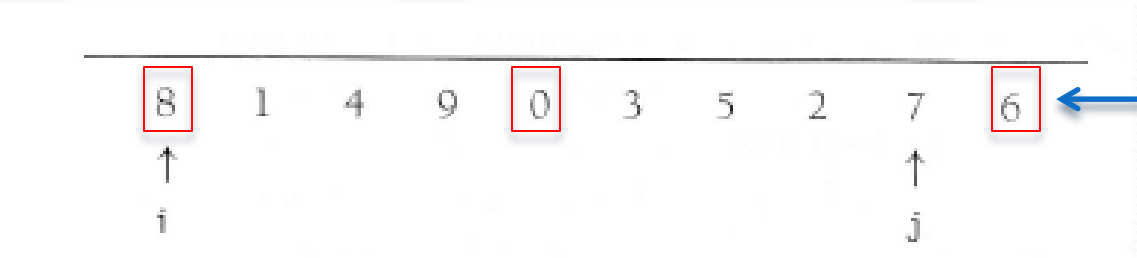
```
// A generic quick-sort type routine. Not the most efficient one.
// Not an in-place implementation.
// Pivot is always the middle item.
template <typename Comparable>
void SortGeneric(vector<Comparable> &items) {
    if (items.size() <= 1) return;

    vector<Comparable> smaller;
    vector<Comparable> same;
    vector<Comparable> larger;

    auto chosen_item = items[items.size() / 2];

    for (auto &x : items) {
       if (x < chosen_item)
           smaller.push_back(std::move(x));
       else if (chosen_item < x)
           larger.push_back(std::move(x));
       else
           same.push_back(std::move(x));
    }

    SortGeneric(smaller); // Recursive call.
    SortGeneric(larger);  // Recursive call.

    // Move from temporary storage to items.
    std::move(begin(smaller), end(smaller), begin(items));
    std::move(begin(same), end(same), begin(items) + smaller.size());
    std::move(begin(larger), end(larger), end(items) - larger.size());
}
```

Picking the pivot

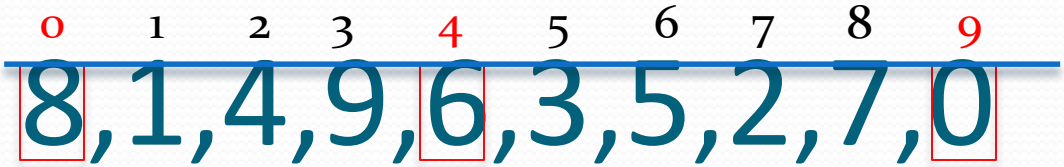
- Choose first element as the pivot?
- Randomly select a pivot?
- Median-of-three pivot selection.
  - Pivot = median $\left(S[0], S\left[\frac{n-1}{2}\right], S[n-1]\right)$
  - E.g.,: 8,1,4,9,6,3,5,2,7,0
  - Pivot = median$(8, 6, 0) = 6$
  - Good for sorted input.



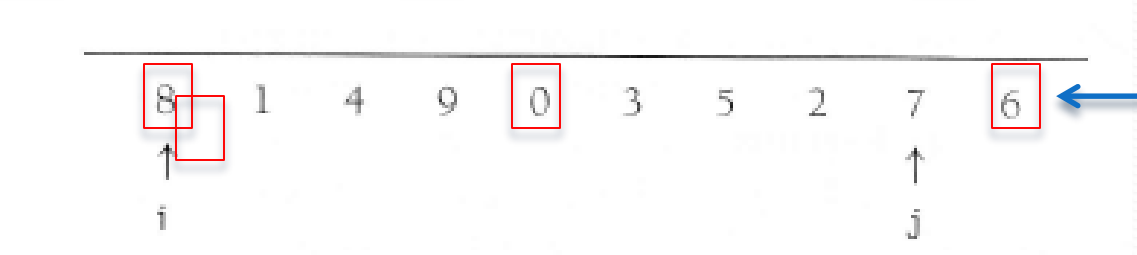
First index: **0**, Last index: **9**,
Middle index: (0+9)/2 = **4**



Compute median-of-3 and swap to end of array



First index: **0**, Last index: **9**,
Middle index: (0+9)/2 = **4**



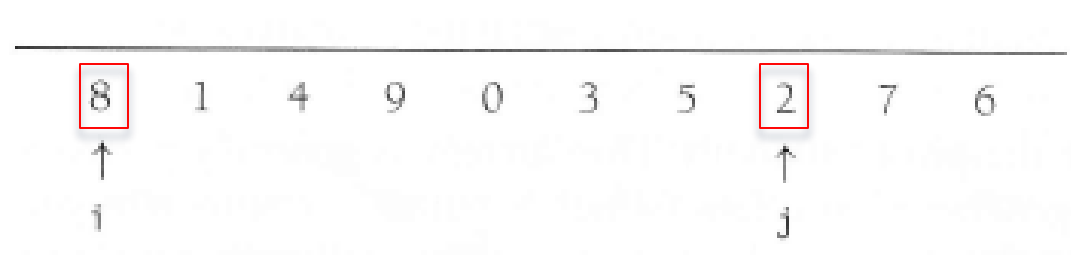
Pivot at end of array

While i < j

move i to the right skipping over elements smaller than pivot

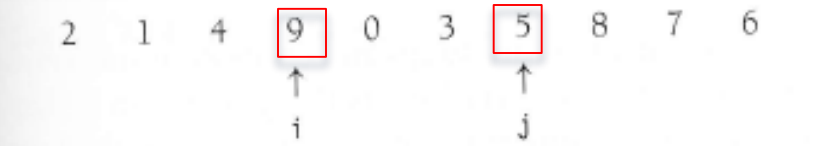
move j to the left skipping over elements larger than pivot

If i<j **swap** a[i] with a[j]

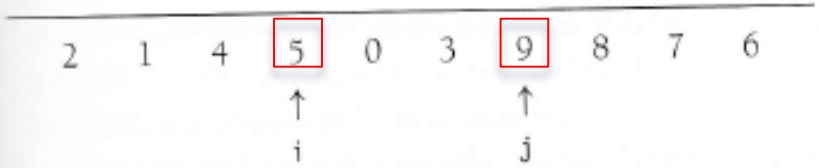


After First Swap

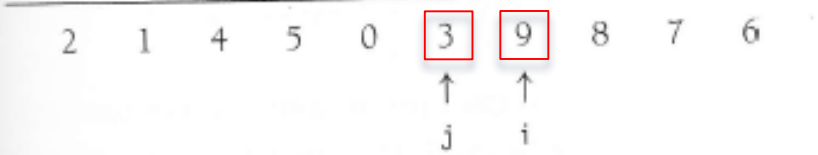
Before Second Swap



After Second Swap

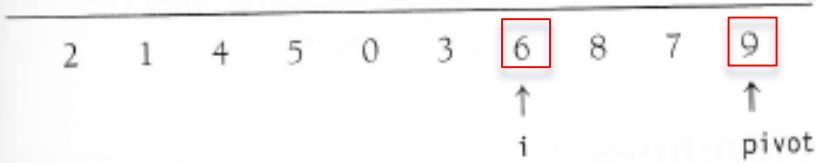


Before Third Swap



At this stage, i and j have crossed, so no swap is performed. The final part of the partitioning is to swap the pivot element with the element pointed to by i.

After Swap with Pivot



While i < j

move i to the right skipping over elements smaller than pivot

move j to the left skipping over elements larger than pivot

If i<j **swap** a[i] with a[j]

Finally, swap a[i] with pivot

Important detail

- How to handle items that are EQUAL to pivot ?
  - Should i stop at element that = the pivot?
  - Should j stop at element that = the pivot?

  - 8 1 6 9 6 3 5 2 7 | 6
    i j

  - 4 4 4 4 4 4 4 4 | 4
    i j

Important detail

- **If both stop:** many swaps, but i, j will cross in middle
  - Good: even partitioning
- If none stop:
  - Prevent them from crossing boundaries
  - i will touch the end, and j will touch the beginning
  - Pivot will swap a[i]
  - Bad: uneven partitioning
- This detail not as silly as it seems.

Important detail

- **If both stop:** many swaps, but i,j will cross in middle
  - Good even partitioning is better for divide-and-conquer
  - $T(n) \approx 2T\left(\frac{n}{2}\right) + n = O(n \log n)$
- If none stop:
  - i will touch the end or j will touch the beginning
  - Pivot will swap a[i]
  - Bad uneven partitioning leads to worst-case behavior
  - $T(n) \approx T(n-1) + n = O(n^2)$

Small arrays

- For N <=20, we can use insertion sort => reduce number of recursive calls.

Analysis of Quicksort

- Running time for array of N elements:

$$T(N) = T(i) + T(N-i-1) + cN, \quad i = |S1|$$

$$T(0) = T(1) = 1$$

Worst-Case

- i = 0 at every step (worst possible partitioning)

$$T(N) = T(0)+T(N-1)+cN \Rightarrow$$
$$T(N) = T(N-1) + cN, \qquad N>1$$

$$T(N-1)=T(N-2)+c(N-1)$$
$$T(N-2)=T(N-3)+c(N-2)$$
$$\dots$$
$$T(2) = T(1) +c(2)$$

==========

$$T(N)=T(1) + c(N+(N-1)+\dots+2)$$

$$\Rightarrow T(N)=\Theta(N^2)$$

Best-Case

- i=N/2 always (best possible partitioning)

  T(N)=T(N/2)+T(N/2)+cN=>

  T(N) = 2T(N/2)+cN

  --------same as mergesort-----→

  T(N)=O(NlogN)

Average-Case

$T(N) = T(i) + T(N-i-1) + cN$, i=size of S1, $N>=2$

$T(0)=T(1)=1$

How do we define average case?--→

Sizes of S1 are equally likely, i.e.

S1 can be of size 0, 1, 2, …, or N-1 with equal probability 1/N:

$Prob\{ |S1| = k \} = 1/N$ for k=0, …, N-1

|A| is the size of set A

Average-Case

$T(N) = T(i) + T(N-i-1) + cN$, i=size of S1, N>=2

$T(0)=T(1)=1$

How do we define average case?--→

It follows that sizes of **S2** are equally likely also, i.e.

S2 can be of size 0, 1, 2, …, N-1 with equal prob 1/N:

$Prob\{ |S2| = k \} = 1/N$ for every k=0, …, N-1

Average-Case

- Let us rewrite:

```
Tavg(|S|) = Tavg(|S1|) + Tavg (|S2|) + cN,
N=|S|

N>=2
```

Average-Case

- **Expected** value of T(i) (|S1|=i) is

  $(1/N)\ T(0)+\ (1/N)\ T(1)+\ldots+\ (1/N)\ T(N-1)=$
  $(1/N)\ (T(0)+\ldots+T(N-1)\ =$
  $(1/N)\sum_{j=0}^{N-1} T(j)$

- The above is the expected value of T(N-i-1) (|S2|=n-i-1) as well

Average-Case

- Recurrence relation becomes:

$$T(N) = (2/N)[\sum_{j=0}^{N-1} T(j)] + cN,\ N>=2$$

or

$$NT(N) = 2\ [\sum_{j=0}^{N-1} T(j)] + cN^2$$

of course this is true for N-1 as well:

$$(N-1)T(N-1) = 2\ [\sum_{j=0}^{N-2} T(j)] + c(N-1)^2$$

Average-Case

- By subtraction:

$$NT(N) - (N-1)T(N-1) = 2\left[\sum_{j=0}^{N-1} T(j)\right] - 2\left[\sum_{j=0}^{N-2} T(j)\right] + cN^2 - c(N-1)^2$$

$$= 2\,T(N-1) + 2cN - c \Rightarrow \text{(-c not significant)}$$

$$NT(N) = (N+1)T(N-1) + 2cN \Rightarrow$$

$$T(N) = ((N+1)/N)\,T(N-1) + 2c \Rightarrow$$

$$\mathbf{T(N)/(N+1) = T(N-1)/N + 2c/(N+1)}$$

Average-case

- $T(N)/(N+1) = T(N-1)/N + 2c/(N+1)$ **[N]**

Telescope:

$T(N-1)/N = T(N-2)/(N-1)+2c/N$ **[N-1]**

$T(N-2)/(N-1)=T(N-3)/(N-2)+2c/(N-1)$ **[N-2]**

....

$T(2)/3 = T(1)/2 +2c/3$ **[2]**

---------- **+all**

$T(N)/(N+1)=T(1)/2 + 2c/(N+1) + 2c/N + \ldots +2c/3$

Average-case

- **$T(N)/(N+1)=T(1)/2 + 2c/(N+1) + 2c/N + \ldots + 2c/3$**

Compute: $2c/(N+1)+\ldots 2c/3=2c*(1/3+1/4+\ldots+1/(N+1))$

$1/3+1/4+\ldots+1/(N+1) = \log_e(N+1)+\gamma-3/2,$

$\gamma \sim 0.577$ (Euler's constant)

- Finally!! $T(N)/(N+1)=O(\log N)$ or

  $T(N)=O(N\log N)$

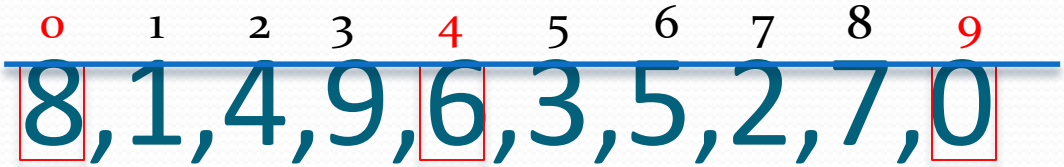
Linear-Expected-Time for Selection

- Selection problem
- Using heap find kth largest (or smallest) in O(N+klogN) time => O(NlogN) for median

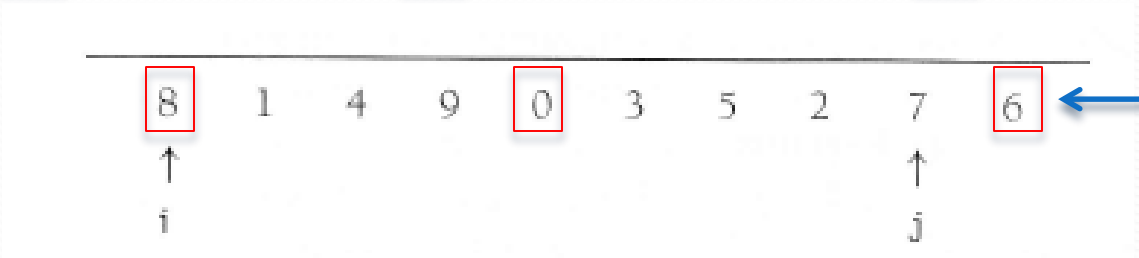
Linear-Expected-Time for Selection

quickselect(S,k)

- If |S|=1, then k=1 and return element of S
  - You can also use CUTOFF for small arrays:
    if |S| <=CUTOFF then sort and return the k_th element
- Pick a pivot v in S
- Partition S-{v} into S1 and S2 as was done in quicksort
- If k<= |S1|, return quickselect(S1, k)
- Else if k==|S1|+1, return pivot (why?)
- Else, return quickselect(S2, k - |S1| -1)

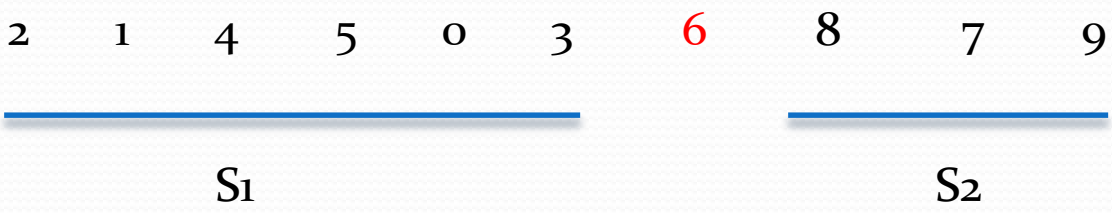


First index: **0**, Last index: **9**,
Middle index: (0+9)/2 = **4**



Compute median-of-3 and swap to end of array

After partitioning:



If k was 7, then solution is pivot (6 is the kth smallest element)
Otherwise, you should only look in S1 or S2 (but not both).
For example if k = 3 you should look in ?
if k = 9 you should look in?

Quick Select analysis

- Worst-case ?
- Average-case?

Quick Select analysis

- Worst-case

$T(N) = T(N-1) + cN,\ N>=2 \Rightarrow ?$

- Average-case?

Quick Select analysis

- Worst-case

$T(N) = T(N-1) + cN,\ N>=2 \Rightarrow O(N^2)$

- Average-case?

Average-Case Quick Select

$$T(|S|) = Tavg(|S1| \text{ or } |S2|) + cN,\ N=|S|$$

$$N>=2$$

Average-Case Quick Select

- Recurrence relation becomes:

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Average-Case Quick Select

- By subtraction:

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$$NT(N) = N\,T(N-1) + 2cN \Rightarrow$$

$$T(N) = T(N-1) + 2c \Rightarrow$$

$$\mathbf{T(N) = O(N)}$$

Quick Select analysis

- Worst-case: $O(N^2)$
- Average-case: $O(N)$

➔ Comparison with heap-based selection algorithm?

Quick Select analysis

- Worst-case: $O(N^2)$

- Average-case: $O(N)$

➔ Note that average case is better than heap-based algorithms for selection!!

Finding Median (i.e. k = N/2) is O(NlogN) for heap-based algorithms.

Introsort

- Introsort or introspective sort is a hybrid sorting algorithm that provides both fast average performance and (asymptotically) optimal worst-case performance. It begins with quicksort, it switches to heapsort when the recursion depth exceeds a level based on (the logarithm of) the number of elements being sorted and it switches to insertion sort when the number of elements is below some threshold.
- In C++, std::sort typically uses introsort.

Introsort – Practical Benefits

- Quicksort is typically the fastest (on average) pure sort.
- Insertion sort is one of the fastest sorts on small inputs since it requires no overhead (no recursive calls, no memory allocation, no setup like buildheap in heapsort, etc.) and is faster than the other "simple" sorts.
- Heapsort is O(n log n) in the worst case, thus we're able to fall back on it for inputs that quicksort performs poorly on.
- All three sorts introsort is composed of are in-place, thus introsort as a whole is in-place.

Introsort – Complexity Analysis

- While quicksort is usually $O(n^2)$ in the worst case, by stopping it after $O(\log n)$ iterations, we keep the complexity of the quicksort portion down to $O(n \log n)$.
- Thus, in the case we need to switch to heapsort, we end up with an $O(n \log n)$ quicksort portion plus an $O(n \log n)$ heapsort portion, for a total of $O(n \log n)$.
- If we don't switch to heapsort, it means we reached small partitions in less than $O(\log n)$ iterations of quicksort, thus still ensuring the total complexity is $O(n \log n)$.

Introsort – Complexity Analysis

- Since we only switch to insertion sort when we reach a range below a constant size, the insertion sort portion is actually O(n). Let's take a look why:
- Consider the threshold c=20, under which we switch to insertion sort. Then, consider an input of size 1,000.
- Since insertion sort is quadratic, the worst case would be 50 partitions of size 20 each.
- Then, the total time taken, approximating insertion sort's worst case at $n^2/2$ operations, is $50*20^2/2$.
- If we had an input size of 1,000,000 instead, we would get 50,000 partitions of 20 each, for a total of $50,000*20^2/2$ operations.

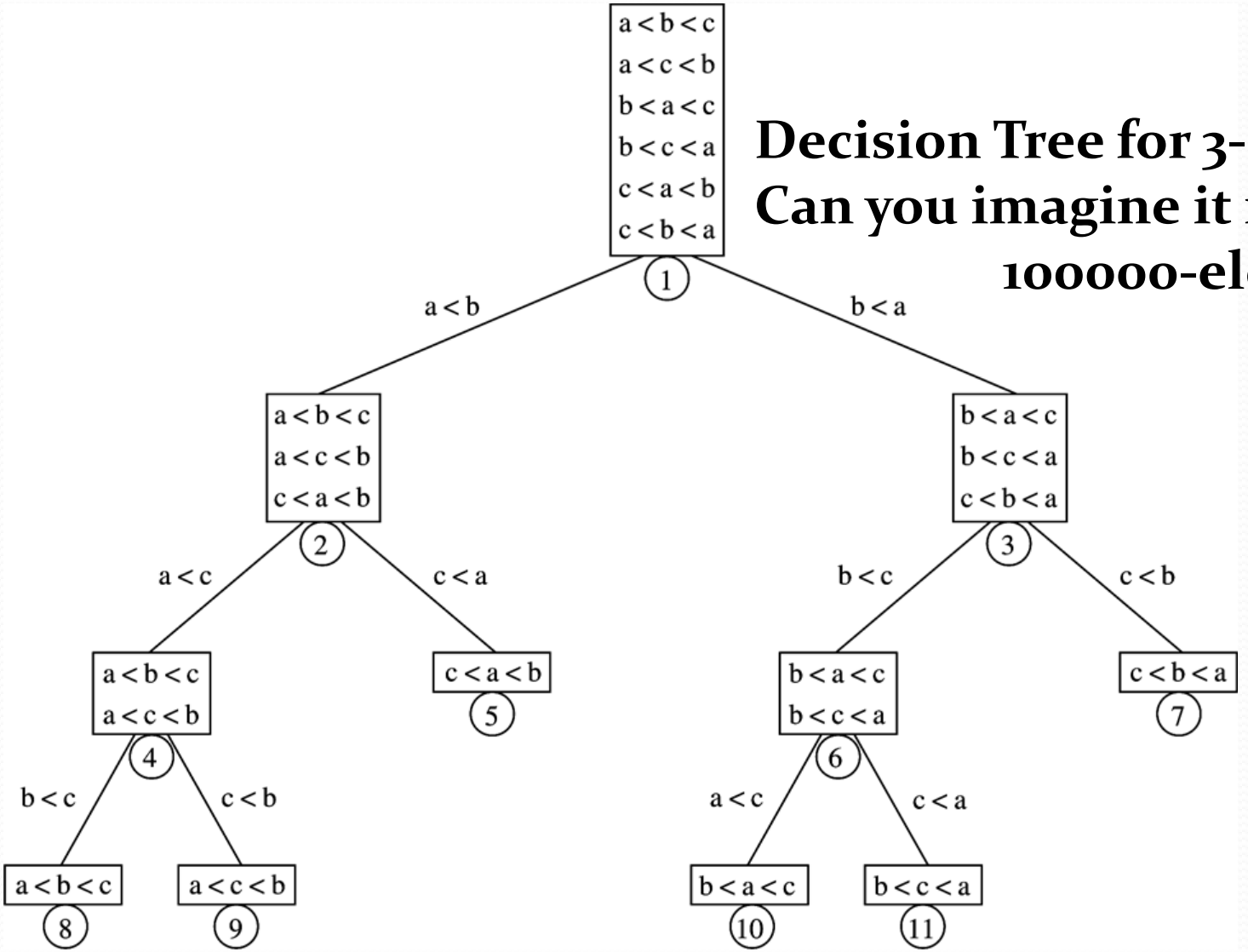
A General Lower Bound for Sorting

- **Theorem:** Any sorting algorithm that uses **only comparisons** requires **$\Omega(N \log N)$** comparisons in the worst case !
- Holds true for **average** case as well.
- What does this mean for mergesort, heapsort, quicksort?

A General Lower Bound for Sorting

- **Theorem:** Any sorting algorithm that uses **only comparisons** requires **Ω(N logN)** comparisons in the worst case !
- Holds true for **average** case as well.
- What does this mean for mergesort, heapsort, quicksort?
- **THEY ARE OPTIMAL!!**

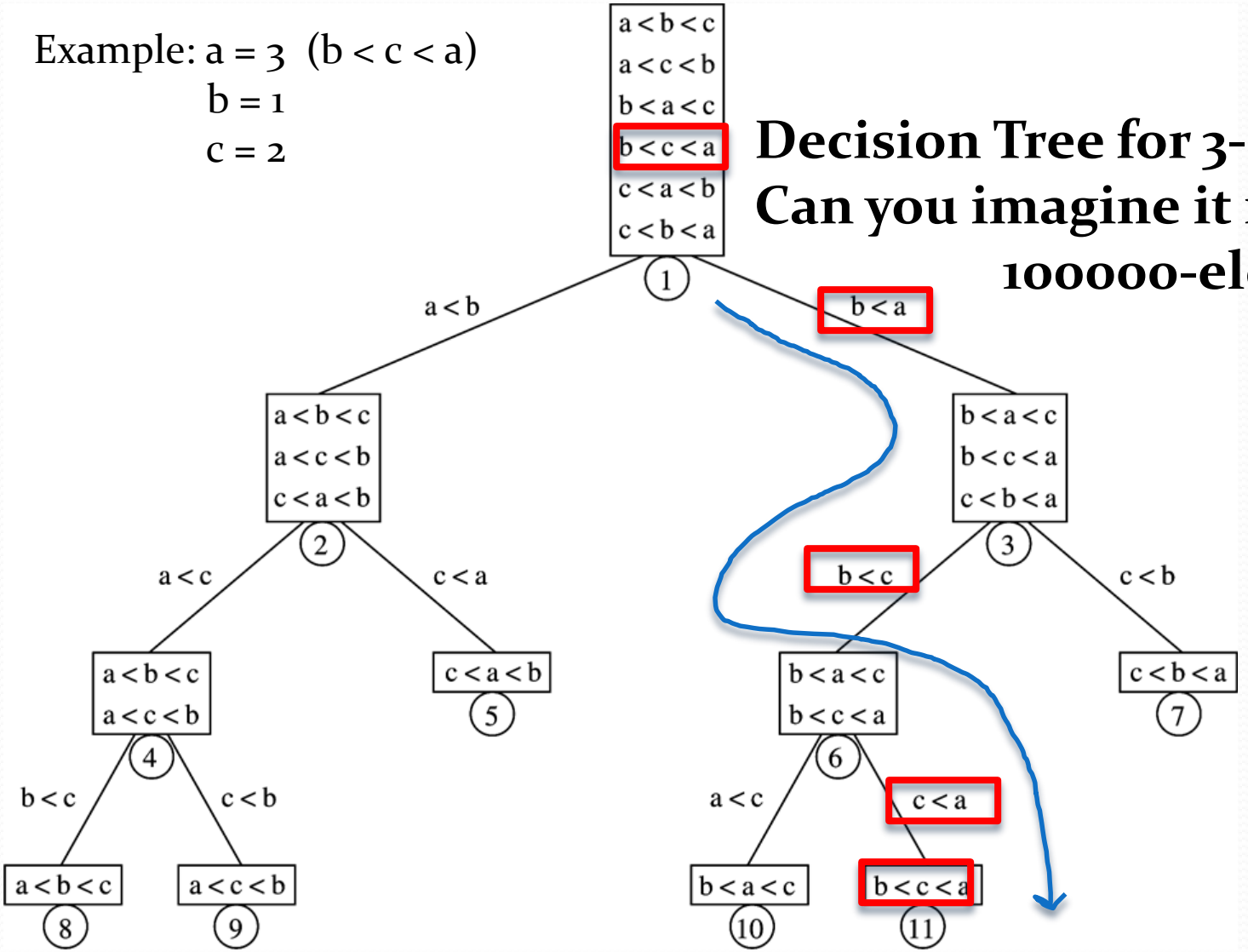
Decision Trees



Decision Tree for 3-element sort
Can you imagine it for
100000-element sort?

Decision Trees

Example: a = 3 (b < c < a)
b = 1
c = 2



Decision Tree for 3-element sort
Can you imagine it for
100000-element sort?

Decision trees

- Every algorithm using comparisons, can be represented by decision tree.
- The length of each path to leaves is the minimum number of comparisons any algorithm needs for a particular input.
- **Information-theoretic bound**
  - Does not give the algorithm to achieve the NlogN bound.
  - It tells you though that this is the **minimum number of comparisons any algorithm will require.**

Proof flow

- Lemma 1: Let T be a binary tree of depth d.
  It has at most $\mathbf{2^d}$ **leaves**.

- Lemma 2: A binary tree of **L** leaves must have
  depth of at least ceil(**logL**),

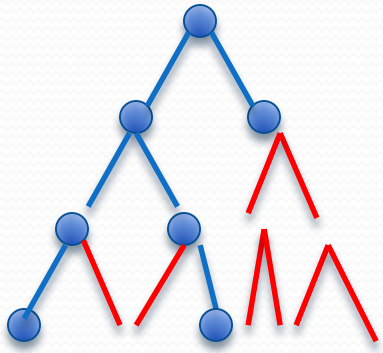
  i.e. d ≥ ceil(logL).

So, if you only know the number of leaves of a tree, you can figure out its minimum depth.

For example, if L = 17, then d >= ceil(log17) = ceil(4.08) =>
d >= 5.

If L = 2^16, then d >=ceil(log2^16) = 16.

Example



Binary tree (blue) of depth d = 3

Full binary tree (blue & red) of depth = 3

Number of nodes of full binary tree:
2^{d+1} = 16

Number of leaves of full binary tree: 2^d = 8

If L is the number of leaves of any tree of depth d

Then L <= 2^d

In this example L (=3) <= 8

Therefore d >= ceil(log L) = ceil(log 3) = ceil(1.5) = 2,
i.e. d >= 2 in this example.

L <= 2^d

d >= ceil(logL)

Proof

Lemma 1. A binary tree of depth d has at most 2^d leaves.

Proof (by induction). If $d = 0$ then the tree has one leaf node, the root. If $d > 0$ then the tree has two subtrees of depth up to $d - 1$ with at most 2^{d-1} leaves, by the inductive hypothesis. Since the root itself is not a leaf this means the number of leaves is at most $2 \cdot 2^{d-1} = 2^d$.

□

Proof flow

Theorem 1:

Any sorting algorithm that is using comparisons between elements requires at least **ceil(log(N!))** comparisons in the **worst case**, i.e.

Number of comparisons ≥ **ceil(log(N!))**

Proof flow

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Any sorting algorithm that is using comparisons between elements requires at least **ceil(log(N!))** comparisons in the **worst case**, i.e.

Number of comparisons ≥ **ceil(log(N!))**

Proof:

A decision tree of N items has N! leaves.

Use Lemma 2 to show that the depth of the decision tree is ≥ ceil(log(N!))

Proof flow

Theorem 2:

Any sorting algorithm that uses only comparisons requires $\Omega(N \log N)$ comparisons in the worst case.

Proof

Theorem 2. Any sorting algorithm that is using comparisons requires $\Omega(n \log n)$ in the worst-case.

Proof. From the previous theorem, $\log(n!)$ comparisons are required.

$$\begin{aligned}
\log(n!) &= \log n + \log(n-1) + \cdots + \log 2 + \log 1 \\
&\geq \log n + \log(n-1) + \cdots \log\left(\frac{n}{2}\right) \\
&\geq \frac{n}{2} \cdot \log\left(\frac{n}{2}\right) \\
&\geq \frac{n}{2}(\log n - \log 2) \\
&\geq \frac{1}{2} n \log n - \frac{1}{2} n \\
&= \Omega(n \log n)
\end{aligned}$$

□

Linear-Time Sorting Algorithms

- Does not only use comparisons to sort.
- Depends on fixed length of inputs
- Can be inefficient for arbitrarily large input values.
- Counting sort, bucket sort, radix sort (4^{th} edition)